



OPEN Sex differences in marathon pacing: analysis of 873,000 Berlin marathon runners reveals men are twice as likely to “hit the wall”

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To determine whether sex differences in marathon pacing strategies and susceptibility to catastrophic deceleration (“hitting the wall”) persist in a massive, high-performance dataset, and to quantify this risk across different performance levels. We analyzed 873,334 finishers from the Berlin Marathon (1999–2025). Pacing stability was quantified using the percentage slowdown in the second half relative to the first. “Hitting the wall” was operationally defined as a deceleration of $\geq 20\%$. A stratified analysis was performed to compare risks between sexes across five performance categories (Competitive to Casual). Multivariable logistic regression adjusted for age and performance category, sensitivity analyses (deduplicated subset; alternative thresholds), and fine-grained pacing metrics from 5 km splits were also conducted. Male runners exhibited significantly greater mean deceleration ($10.73\% \pm 11.41\%$) compared with female runners ($8.34\% \pm 8.91\%$, $p < 0.001$). The prevalence of “hitting the wall” was nearly double in men (17.63%) compared with women (9.66%), corresponding to a crude Odds Ratio of 2.00 (95% CI 1.97 to 2.03); after adjustment for age and performance category, the disparity strengthened (adjusted OR = 3.88, 95% CI 3.81 to 3.94). The risk disparity widened among the fastest runners: in the Competitive (< 3 h) category, male runners were approximately six times more likely to experience catastrophic deceleration than their female counterparts (1.42% vs 0.23%). The gap was stable across the 27-year archive (Mann–Kendall $\tau = 0.14$, $p = 0.33$). Despite faster finish times, men demonstrate significantly less stable pacing strategies and a twofold higher crude risk of catastrophic deceleration compared with women, with the disparity most pronounced among the fastest runners. These findings are consistent with the hypothesis that behavioral and strategic factors contribute alongside physiological determinants to sex differences in marathon outcomes, and provide a quantitative basis for further investigation of the underlying mechanisms.

Keywords Marathon, Pacing strategies, Sex characteristics, Athletic performance, Fatigue

The marathon has evolved from a niche elite discipline to a global mass-participation phenomenon^{1,2}. Unlike shorter endurance events where performance is predominantly limited by maximal oxygen uptake (VO_2max) and power output, the marathon imposes a unique physiological constraint: the finite limit of glycogen stores³. Consequently, success in the marathon is not solely defined by “how fast one can run,” but rather by “how efficiently one can manage energy” over 42.195 km⁴. This distinct characteristic makes pacing strategy—the distribution of work rate over time—the most critical tactical determinant of performance.

Historically, male runners have consistently outperformed female runners in absolute finish times, a disparity largely attributed to well-documented physiological advantages such as greater muscle mass, lower body fat percentages, greater cardiac mass, higher blood volume, and higher hemoglobin concentrations^{5–7}. However, absolute speed does not equate to pacing efficiency. Emerging evidence from sports psychology and behavioral economics suggests a paradox: while men are physiologically faster, they may be behaviorally less efficient⁸. Theoretical models of decision-making under fatigue propose that men may display higher tendencies toward “risk-taking behavior” and “overconfidence,” which could contribute to aggressive initial splits that often prove

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unsustainable⁹. In contrast, female runners are frequently hypothesized to possess superior self-pacing abilities and greater resistance to decision-making fatigue¹⁰.

Despite the importance of this phenomenon, the existing literature on sex differences in marathon pacing remains inconclusive. Previous studies have often been limited by small sample sizes ($n < 1,000$), inconsistency in environmental conditions, or an exclusive focus on elite athletes who do not represent the general running population^{11–13}. A critical unresolved question remains: Is the male tendency to decelerate a biological inevitability, or is it a behavioral pattern hypothesized to relate to competitive risk-taking dynamics that may persist across all levels of ability? To answer this, a large-scale, “Big Data” approach is required to analyze pacing stability across the entire spectrum of human performance, from the elite sub-3 h runner to the recreational finisher.

The primary objective of this study was to analyze the pacing strategies and performance resilience of over 870,000 finishers at the Berlin Marathon (1999–2025). By utilizing a dataset from a World Marathon Major known for its flat profile and consistent conditions, we aimed to isolate sex as a variable, minimizing topographic confounders. We hypothesized that male runners, regardless of their performance level (Competitive to Casual), would exhibit significantly more aggressive pacing strategies (positive splits) and a higher prevalence of catastrophic deceleration (“hitting the wall”) compared to female runners.

Materials and methods

Study design and data source

This study employed a retrospective observational design leveraging a large-scale dataset to analyze pacing strategies and performance resilience among marathon runners. Data were extracted from the official BMW Berlin Marathon Results Archive, covering a 27-year period from 1999 to 2025. The Berlin Marathon was selected as the primary data source due to its status as a World Athletics Platinum Label Road Race and a World Marathon Major. Its predominantly flat course profile and historically consistent environmental conditions minimize the confounding effects of topography on pacing behavior, establishing an ideal setting for investigating the physiological and behavioral dynamics of endurance running.

The dataset consists of official race results recorded via electronic transponder timing systems (chip timing). For each participant, the primary variables retrieved included net finish time (duration from crossing the start line to the finish line), half-marathon split time (21.0975 km), 5 km split intervals (where available and consistent across years), sex, and age group categorization. Given that the data were obtained from a publicly accessible repository and analyzed in an anonymized format, this study was exempt from Institutional Review Board (IRB) approval, in accordance with standard ethical guidelines for the use of public secondary data.

Data pre-processing and normalization

The initial dataset comprised a total of 880,779 raw records extracted from the archive (1999–2025). Prior to statistical analysis, these data underwent a systematic pre-processing and normalization procedure to ensure consistency across the 27-year longitudinal archive. Categorical variables, specifically sex, were harmonized from various archival formats—including distinct German and English terminologies used in earlier editions of the race—into a standardized binary classification. Temporal variables, originally recorded in string format (hours:minutes:seconds), were converted into continuous numeric values (total seconds) to facilitate the mathematical computation of derived pacing metrics. To ensure analytical rigor and physiological validity, the dataset was filtered to exclude biologically implausible performances ($< 1:59:00$) and recreational walking efforts exceeding the official event time limit ($> 6:15:00$). Additionally, records containing missing values for critical pacing checkpoints—specifically the final net time—were removed via listwise deletion. Consequently, 7,445 records (0.85% of the initial sample) were excluded, resulting in a complete-case dataset of 873,334 finishers for the subsequent analysis.

The dataset constitutes race entries rather than uniquely identified runners; the same individual may appear in multiple editions across the 27-year archive. The official BMW Berlin Marathon Results Archive does not provide a persistent runner identifier; therefore, to assess the potential impact of non-independent observations, a sensitivity analysis was conducted on a deduplicated subset constructed by retaining the first appearance per composite key of normalized runner name (lowercased, whitespace- and accent-stripped) and age group. This conservative approach may underestimate true uniqueness due to spelling variations across editions and is treated as a robustness check rather than a definitive correction. The deduplicated subset comprised $n = 700,877$ entries (80.3% of the pacing-valid analytical cohort of 872,670; first appearance per composite key), and results from this sensitivity analysis are reported in Supplementary Table S3.

Operational definitions of pacing metrics

To quantify pacing behavior independent of absolute performance levels, a dimensionless Pacing Index (PI) was calculated for each participant. The PI is defined as the ratio of the time taken to complete the second half of the marathon (21.0975 km to 42.195 km) to the time taken for the first half (Start to 21.0975 km). To enhance interpretability, this ratio was converted into a Percentage Slowdown metric, calculated as $(PI - 1) \times 100$. In this framework, a value of 0% represents an even split (constant velocity), while positive values indicate a “positive split” (deceleration) and negative values indicate a “negative split” (acceleration).

Furthermore, to distinguish between normative physiological fatigue and catastrophic pacing failure, “hitting the wall” was operationally defined as a percentage slowdown of $\geq 20\%$ in the second half relative to the first. This threshold was selected based on metabolic limitations described in exercise physiology literature. The threshold corresponds to the magnitude of deceleration expected during the transition from carbohydrate- to lipid-dominant oxidation, where the lower rate of ATP resynthesis per unit oxygen for lipids relative to carbohydrates

is expected to reduce sustainable power output by approximately 15–25%. We use the threshold as an operational proxy for catastrophic pacing decline and do not interpret it as direct evidence of glycogen depletion.

This high threshold distinguishes minor fatigue-induced deceleration from catastrophic pacing failure, allowing for a binary risk assessment across different demographic groups. The 20% criterion has also been employed in previous large-scale marathon studies^{14,15}; a related threshold defined over cumulative 5 km segments was used by Smyth in an analysis of more than 4 million marathon performances¹⁶. To verify robustness, sensitivity analyses were conducted at alternative thresholds of 15% and 25% (Supplementary Table S2). Additionally, a graded severity classification (mild 10–15%, moderate 15–20%, severe 20–25%, catastrophic > 25%) is reported in Supplementary Table S2o to characterize the continuum of pacing decline by sex.

Performance stratification

To control for the well-established correlation between absolute performance level and pacing stability—where faster runners typically exhibit more even pacing strategies—the cohort was stratified into five distinct performance categories based on net finish time. This stratification ensures that comparisons between sexes are conducted within groups of similar physiological capability, mitigating the confounding effect of fitness level on pacing behavior. The categories were defined as follows: Competitive (finish time < 3:00 h), Advanced (3:00–3:30 h), Intermediate (3:30–4:00 h), Recreational (4:00–4:30 h), and Casual (> 4:30 h). This segmentation facilitates a granular analysis, allowing the study to determine whether observed sex disparities in pacing and resilience persist uniformly across the spectrum of amateur and sub-elite performance.

To address potential confounding from age- and sex-specific competitive context, a complementary sensitivity analysis was conducted by re-classifying runners into quintiles of finish time computed within each (sex, age group) stratum (Q1 = top 10%, Q2 = 10–25%, Q3 = 25–50%, Q4 = 50–75%, Q5 = bottom 25%). The main analyses (incidence of “hitting the wall” and Odds Ratios) were re-computed under this within-cohort percentile classification; results are reported in Supplementary Table S4. Together with the multivariable logistic adjustment described in §Statistical Analysis, this provides complementary demographic adjustment alongside the absolute-time stratification reported in Table 2.

Statistical analysis

Descriptive statistics were computed to summarize central tendencies (mean, median) and dispersion (standard deviation, interquartile range) for all pacing metrics. The assumption of homogeneity of variance between sex groups was assessed using Levene’s test. Given the large sample size and the results indicating unequal variances ($p < 0.001$), comparisons of mean Percentage Slowdown were conducted using Welch’s *t*-test, which does not assume homoscedasticity. Additionally, the non-parametric Mann–Whitney *U* test was employed to compare the distributions of pacing strategies, providing robustness against the skewness observed in the data.

Effect sizes were quantified using Cohen’s *d* to determine the magnitude of differences, classified as small ($d \geq 0.2$), medium ($d \geq 0.5$), or large ($d \geq 0.8$). To analyze the incidence of “hitting the wall,” a Chi-Square (χ^2) test of independence was performed to evaluate the association between sex and pacing failure. Risk quantification was further expressed through Odds Ratios (OR) with 95% confidence intervals.

To control for potential confounding by age and performance level, a multivariable logistic regression model was additionally fitted with “hitting the wall” as the binary outcome, and sex (factor; reference: female), age (continuous, mid-point of the 5-year age group), and performance category (5-level factor; reference: Casual) as predictors. A sex $\times$ age interaction term was tested to assess effect modification. Adjusted Odds Ratios with 95% confidence intervals are reported in Supplementary Table S1.

Logistic model diagnostics were computed on the analytical cohort ($n = 855,061$ logistic-valid finishers; 17,609 records excluded from the $n = 872,670$ pacing-valid sample for missing performance-category or age-group label, see Supplementary Table S6 Panel F for the exclusion cascade) to address concerns about multicollinearity, goodness-of-fit, calibration, nonlinear age effects, and influential observations (Supplementary Table S6). Multicollinearity was assessed by Variance Inflation Factors for each predictor; goodness-of-fit by McFadden’s pseudo- R^2 , AIC and BIC for both the main and interaction models; calibration by a tenfold cross-validated recalibration intercept and slope, complemented by a decile calibration table and the Hosmer–Lemeshow chi-square statistic (interpreted with caution given the very large sample size); the linearity assumption for age was tested by likelihood-ratio comparisons against a quadratic and a natural-cubic-spline parameterization (3 knots placed at the 10th, 50th, and 90th quantiles of the age-midpoint distribution, with a sum-to-zero constraint); influential observations were summarised via the Pregibon *dbeta* approximation. Given the extremely large sample size, statistical significance is reached for virtually all comparisons; interpretation throughout the Results section therefore prioritises effect magnitude and absolute prevalence differences over *p*-values alone.

For the subset of runners with complete 5 km splits ($n = 856,759$, 98.1%), three fine-grained pacing metrics were computed per runner: (1) the coefficient of variation (CV) of pace across all 5 km segments, capturing overall pacing irregularity; (2) the inflection point, defined as the first 5 km segment at which pace exceeded the runner’s pre-half-marathon mean by more than 5% and remained above that threshold for all subsequent segments through the finish; and (3) late-race deceleration, defined as the percentage difference between pace in the 35–40 km segment and the 5–10 km segment. Two additional exploratory metrics—oscillation count (split-to-split sign changes in pace gradient) and km30 gradient (slope of pace evolution near the typical wall onset)—were computed and are reported in Supplementary Table S5. Metrics were compared by sex using Welch’s *t*-tests and Mann–Whitney *U* tests. The 5% deviation threshold was selected to exceed the typical within-runner coefficient of variation of pace across consecutive 5 km segments observed in even-paced marathon runners (approximately 3–5% in mass-participation cohorts; cf.^{16,17}), thereby identifying deviations that are unlikely to reflect spontaneous pacing fluctuation. The requirement that pace remain above this threshold for all subsequent segments through the finish was imposed to distinguish a sustained metabolic-style inflection from

transient mid-race pace excursions (e.g., aid-station decelerations or terrain-induced variability). Sensitivity to alternative thresholds (3% and 7%) was not formally tested in the present analysis and represents an avenue for future trajectory-based work.

The temporal evolution of the sex disparity in wall-hit prevalence over the 27-year archive (1999–2025) was assessed using the Mann–Kendall trend test^{18,19} on the annual prevalence series, complemented by linear regression with year as predictor of the prevalence gap (Fig. 5). The 2020 edition was cancelled due to COVID-19 and is therefore absent from the temporal analysis (*n* = 26 editions).

Statistical significance was established at an alpha level of *p* < 0.05. All data processing, statistical inference, and visualization were performed using the Python programming language (version 3.12). Key libraries included Pandas and NumPy for data manipulation, SciPy for statistical testing, and Matplotlib and Seaborn for the generation of high-resolution figures. To promote reproducibility and open science, the complete analytical code, detailed statistical reports, and supplementary figures are available in a public repository.

Results
Participant characteristics

The final analytical cohort comprised *n* = 873,334 finishers from the Berlin Marathon (1999–2025). As summarized in Table 1, the demographic distribution was predominantly male, with men accounting for 75.5% (*n* = 659,294) of the sample, while women constituted 24.5% (*n* = 214,040).

Performance analysis revealed significant differences in absolute finish times. Male runners achieved a mean net finish time of 4:02:22 (± 00:42:36), whereas female runners recorded a mean time of 4:28:36 (± 00:41:28). To account for the skewness typical of mass-participation endurance events, median times and interquartile ranges (IQR) were analyzed. The median finish time for men was 3:57:46 (IQR: 3:32:02–4:28:49), indicating that the core 50% of the male cohort finishes within a ~56-min window. In contrast, the median finish time for women was 4:26:02 (IQR: 3:58:20–4:55:40), with a slightly wider central dispersion.

Regarding age distribution, the cohort displayed a mature profile characteristic of marathon distances. The 40-age category (representing runners aged 40–44) was the most prevalent, comprising 162,234 runners (~18.6% of the cohort). This was closely followed by the 35 category (*n* = 143,728; ~16.5%) and the 45 category (*n* = 142,610; ~16.3%), suggesting that over 50% of the field is concentrated between the ages of 35 and 49.

Global analysis of pacing strategies (macro view)

Descriptive analysis revealed a distinct sex disparity in pacing conservation across the entire cohort. Men exhibited a significantly higher mean Percentage Slowdown (10.73% ± 11.41%) compared to women (8.34% ± 8.91%), resulting in a mean performance gap of 2.39 percentage points. The larger standard deviation observed in the male cohort further indicates greater variability and inconsistency in pacing execution compared to the female cohort.

Beyond central tendency, the distribution of pacing strategies exhibited marked asymmetry. While both sexes displayed positive skewness—indicating a general tendency toward deceleration rather than acceleration (negative splits)—the male distribution showed a higher degree of skewness (1.59) compared to the female distribution (1.35). This statistical characteristic confirms that the male cohort possesses a heavier “right tail,” representing a higher prevalence of extreme positive splits and catastrophic pacing failures.

Inferential testing confirmed these observations were statistically robust. Levene’s test indicated unequal variances between groups (*p* < 0.001), justifying the use of Welch’s *t*-test, which yielded a statistically significant difference between mean slowdowns (*p* < 0.001). Furthermore, the non-parametric Mann–Whitney *U* test confirmed that the distribution of male pacing strategies was stochastically “slower” (i.e., greater deceleration) than that of females (*p* < 0.001). The effect size for this global difference was calculated as *d* = 0.22, interpreted as a small but biologically relevant effect given the context of endurance performance, where marginal percentage differences equate to substantial time deficits over the marathon distance.

Incidence and risk of “hitting the wall”

To investigate susceptibility to catastrophic fatigue, the study analyzed the incidence of “hitting the wall,” operationally defined as a pacing deceleration of ≥ 20% in the second half of the marathon. The analytical sample for this specific metric comprised *n* = 872,670 runners, excluding a minor subset of 664 finishers (0.08% of the total cohort) who lacked valid half-marathon split data required for the calculation of the Pacing Index.

Characteristic	Men	Women	Total/average
Sample size (<i>n</i>)	659,294 (75.5%)	214,040 (24.5%)	873,334
Finish time			
Mean ± SD	4:02:22 ± 42 min	4:28:36 ± 41 min	4:08:48 ± 44 min
Median (IQR)	3:57:46 (3:32–4:28)	4:26:02 (3:58–4:55)	4:03:52 (3:37–4:36)
Age distribution			
Most prevalent group	40–44 (18.6%)	40–44 (18.6%)	40–44
Second prevalent group	35–39 (16.5%)	35–39 (16.5%)	35–39

Table 1. Participant demographics and performance characteristics of the Berlin Marathon cohort (1999–2025). IQR = Interquartile Range (25th–75th percentile). Time format is h:min:s

Analysis of this binary outcome revealed a marked disparity in prevalence between sexes. Within the male pacing cohort ($n=658,790$), a total of 116,150 runners reached this threshold, corresponding to an incidence rate of 17.63%. In contrast, the female cohort ($n=213,880$) demonstrated significantly greater resilience, with only 20,667 runners (9.66%) exhibiting a comparable degree of performance decline.

A Chi-Square test of independence confirmed that the observed association between sex and pacing failure was statistically significant ($\chi^2=7,753.39, p<0.001$). To quantify the magnitude of this risk, an Odds Ratio (OR) was calculated. The analysis yielded an OR of 2.00 (95% CI: [1.97–2.03]), indicating that male runners are exactly twice as likely to experience a catastrophic pacing failure compared to female runners. This finding suggests that for every woman who “hits the wall” in the Berlin Marathon, two men experience the same outcome, highlighting a fundamental difference in physiological or behavioral resilience under race conditions.

After adjusting for age and performance category in a multivariable logistic regression, the sex disparity persisted and was substantially larger in magnitude (adjusted OR=3.88, 95% CI [3.81–3.94], $p<0.001$; Supplementary Table S1), indicating that the crude OR is partly attenuated by the differential demographic mix between male and female cohorts. The sex×age interaction was statistically significant but of small magnitude (interaction OR=0.993 per year, $p<0.001$), suggesting that the male disadvantage attenuates only marginally with increasing age. A sensitivity analysis on the deduplicated subset ($n=700,877$; Supplementary Table S3) confirmed direction and magnitude with virtually unchanged crude OR (1.999, 95% CI [1.97–2.03]), indicating robustness against repeated participation. Full logistic diagnostics—variance inflation factors, calibration assessment, comparison of linear/quadratic/spline parameterisations of age, and influence summary—are reported in Supplementary Table S6.

Stratified analysis by performance level (micro view)

To control for the confounding effect of absolute fitness on pacing strategy, the cohort was stratified into five distinct performance categories based on finish time. As detailed in Table 2, the sex disparity in pacing conservation was robust and persistent across every performance tier ($p<0.001$ for all subgroups).

In the Competitive category (<3:00 h), where physiological conditioning and race strategy are expected to be optimized, male runners still exhibited a significantly higher mean percentage slowdown (3.79%) compared to their female counterparts (2.74%), resulting in a pacing gap of 1.05 percentage points. Notably, while the absolute prevalence of “hitting the wall” was low in this high-performance group, the relative risk was disproportionately high: male runners in the Competitive (<3 h) category were 6.06 times more likely to experience catastrophic deceleration than female runners in the same category (1.42 vs. 0.23%).

As performance level decreased, the magnitude of the sex-stratified slowdown gap widened progressively. In the Recreational category (4:00–4:30 h), the mean performance gap expanded to 4.98 percentage points. In the Casual category (>4:30 h), the sex disparity peaked, with men slowing down by an average of 18.90% compared to 11.85% for women ($d=0.57$). Furthermore, the absolute incidence of pacing failure in this category was severe, with nearly half of all male runners (41.62%) hitting the wall, compared to 17.66% of women.

Fine-grained pacing analysis

For the subset of runners with complete 5 km splits ($n=856,759$; 98.1% of the analytical cohort), male runners exhibited significantly greater pacing irregularity than female runners across all examined metrics. The coefficient of variation (CV) of pace across 5 km segments was higher in men than in women (CV: M=0.077 vs F=0.062; Cohen’s $d=0.27, p<0.001$). The inflection point—the first 5 km segment for which pace exceeded the pre-half-marathon mean by more than 5% and remained slower for all subsequent segments through the finish—occurred slightly later in men (median 25 km, IQR 20–30) than in women (median 20 km, IQR 20–30); however, ~36% of male runners and ~52% of female runners did not exhibit a defined inflection point under this

Performance category	Sample (N) (men/women)	Avg slowdown (%) (Men/Women)	Slowdown gap (pp)	Wall hit rate (%) (Men/Women)	Prevalence ratio (M:F)	Crude OR (M vs F)	95% CI
1. Competitive (<3 h)	40,212/2,131	3.79%/2.74%	+1.05	1.42%/0.23%	6.06	6.14	(2.54, 14.81)
2. Advanced (3:00–3:30)	113,830/12,203	6.02%/3.38%	+2.65	4.61%/0.81%	5.69	5.91	(4.84, 7.22)
3. Intermediate (3:30–4:00)	195,231/43,146	7.47%/4.14%	+3.34	8.08%/1.50%	5.37	5.75	(5.31, 6.23)
4. Recreational (4:00–4:30)	150,163/57,787	11.70%/6.72%	+4.98	18.81%/4.32%	4.36	5.13	(4.92, 5.36)
5. Casual (>4:30)	159,354/98,613	18.90%/11.85%	+7.04	41.62%/17.66%	2.36	3.32	(3.26, 3.39)

Table 2. Stratified analysis of pacing metrics, sex-based slowdown gap, and risk of hitting the wall by performance level. Values are presented as Male/Female. The sex-based slowdown gap represents the difference in percentage points between male and female mean slowdown. Prevalence Ratio is computed as the ratio of male wall-hit prevalence to female wall-hit prevalence within each performance category. Crude Odds Ratios (M vs F) from Wald-type 95% confidence intervals are computed within each performance category from the 2×2 sex×wall-hit cross-tabulation; these stratum-specific Odds Ratios all exceed the crude marginal Odds Ratio (2.00; Results §Incidence) and are internally consistent with the multivariable adjusted estimate (OR=3.88; Table S1), illustrating the negative confounding by performance category (see Discussion for explanation of the negative confounding pattern). Adjusted Odds Ratios from the multivariable logistic regression are reported in Supplementary Table S1. All ratios are statistically significant at $p<0.001$ by Chi-square.

strict sustained-deterioration criterion, reflecting the proportion who maintained relatively even pacing through to the finish. Late-race deceleration (pace 35–40 km vs 5–10 km) showed the largest sex effect ($M = +17.8\%$ vs $F = +12.6\%$ slower in the late segment; $d = 0.31$, $p < 0.001$), indicating that men sustain a substantially larger pace decline in the final stages of the race. Two additional exploratory metrics (oscillation count and km30 gradient) are reported in Supplementary Table S5; the oscillation count showed an unexpected direction (women slightly higher than men, $d = -.18$) and warrants further investigation.

Temporal evolution of the sex disparity in pacing

The 27-year archive (1999–2025; 26 editions analyzed, with 2020 cancelled due to COVID-19) was used to assess whether the sex disparity in wall-hit prevalence has changed over time (Fig. 5). The Mann–Kendall trend test indicated no statistically detectable change in the prevalence gap ($\tau = 0.14$, $p = 0.33$; mean gap = 8.4 percentage points, range 2.8–19.9), supported by linear regression with year as predictor (slope = +0.12 percentage points/year, 95% CI ± 0.17 , $p = 0.17$, $R^2 = 0.08$). Wall-hit prevalence among female runners showed a modest increasing trend ($\tau = 0.34$, $p = 0.02$), while the male series trended upward without reaching statistical significance ($\tau = 0.26$, $p = 0.06$). The high inter-annual variability in the gap (range 2.8–19.9 percentage points) likely reflects environmental and cohort composition differences across editions; the absence of a systematic temporal trend supports the interpretation that the sex disparity is a structural feature of marathon pacing rather than an artifact of any specific era of training norms or qualification policies.

Sensitivity analyses

The robustness of the main findings was further evaluated through complementary sensitivity analyses. First, re-classification by within-cohort sex $\times$ age-group quintiles (Supplementary Table S4) yielded Odds Ratios in the same direction across all five competitive levels (range 1.87–3.53, all 95% CIs strictly excluding 1), indicating that the sex gap is not an artifact of differential age-stratified competitive composition. Second, alternative thresholds for “hitting the wall” (15% and 25% slowdown; Supplementary Table S2) preserved the direction and approximate magnitude of the sex disparity, supporting the robustness of the 20% operational definition. Third, a graded severity classification (mild 10–15%, moderate 15–20%, severe 20–25%, catastrophic > 25%; Supplementary Table S2) revealed that the sex gap persists across the continuum of pacing decline, not only at the binary threshold.

Discussion

The primary aim of this study was to determine whether the sex differences in marathon pacing strategies observed in smaller cohorts persist within a massive, high-performance dataset. Our analysis of over 870,000 finishers confirms that male runners exhibit significantly less stable pacing strategies than female runners, validating our hypothesis that men, regardless of performance level, are more prone to aggressive pacing and catastrophic deceleration. Despite a faster mean finish time, men demonstrated a twofold higher risk (OR = 2.00) of “hitting the wall” compared to women. This finding corroborates previous observations by Deaner et al.²⁰ and Hubble & Zhao¹⁵, who identified similar trends in US-based marathons. However, the unprecedented scale of the current study ($n = 873,334$) provides conclusive evidence that this is a pervasive phenomenon across the global running population, robust against variations in course topography or annual weather conditions^{14,21}.

Perhaps the most counterintuitive finding of this investigation is the amplification of the sex risk disparity among the highest-performing athletes, revealing that superior physiological conditioning does not inoculate male runners against catastrophic pacing failures. It is often assumed that pacing stability improves linearly with performance level and experience²². However, our stratified analysis reveals a paradox: while the *absolute* incidence of hitting the wall decreases with speed, the *relative risk* disparity between sexes actually widens. In the Competitive category (< 3:00 h), male runners were 6.06 times more likely to experience catastrophic deceleration than their female counterparts. This contradicts the notion that pacing errors are solely a function of inexperience. Instead, it suggests that high-performance male runners may be prone to adopting high-risk strategies—running closer to their physiological ceiling—potentially shaped by competitive pressures and the complex dynamics of decision-making under physical stress²³.

Consistent evidence indicates that men exhibit, on average, greater skeletal muscle mass, a larger cross-sectional area of muscle fibers—particularly type II fibers—greater maximal strength, higher muscle power, higher VO_2max and greater anaerobic capacity compared with women^{7,24}. These differences are predominantly attributed to greater lifetime exposure to testosterone during and after puberty, which promotes muscle hypertrophy, a higher proportion of fast-twitch fibers, greater cardiac dimension, blood volume, hemoglobin mass and enhanced glycolytic capacity, resulting in superior performance in tasks requiring high force, power output and aerobic capacity^{7,25}.

Conversely, women showed to present superior pacing stability, which is likely supported by distinct physiological mechanisms. Research into substrate utilization has consistently demonstrated that women exhibit higher rates of lipid oxidation and lower respiratory exchange ratios²⁶ during sub-maximal endurance exercise^{27,28}, and greater relative proportion of type I muscle fibers compared to men^{7,24}. These differences have been associated with differences in muscle fiber typology, greater oxidative capacity, and hormonal modulation by estradiol^{28,29}.

These differences in muscle fiber typology and metabolic profile allows for greater glycogen sparing, effectively delaying the onset of glycogen-depletion-induced fatigue, commonly known as “the wall”³⁰. Consequently, even when running at comparable relative intensities, female runners present a more favourable substrate-utilization profile, which may contribute to the lower prevalence of pronounced second-half deceleration observed in the female cohort²⁸.

Beyond physiological mechanisms, gender-related factors may also contribute to the observed sex disparity in pacing. The behavioral economics literature suggests that men, on average, more often overestimate their competitive ability and tolerate higher risk in competitive contexts^{31–33}, which in a self-paced endurance setting could plausibly manifest as more aggressive starting splits and reduced pacing discipline^{16,17}. The present study, however, did not collect the data required to test these mechanisms directly: psychometric assessment of competitive risk preference, pacing intention and confidence calibration, or qualification-pressure proxies. The behavioral interpretation is therefore presented as a hypothesis to be evaluated by future studies that pair archival pacing data with individual-level psychometric measurement.

Therefore, current evidence is consistent with the hypothesis that, although sex-specific physiological characteristics modulate the development of fatigue during prolonged endurance exercise, behavioral and strategic factors related to pacing decisions may contribute substantially to sex differences in marathon pacing^{16,17}. However, the present dataset does not directly measure psychological variables, and alternative mechanisms—including competitive density, qualification pressure, tactical effects, substrate utilization, training distribution, and participation demographics—warrant further investigation⁷.

The substantial increase from the crude (OR = 2.00) to the adjusted (OR = 3.88) estimate of the male-to-female odds of hitting the wall reflects negative confounding by performance category. In this cohort, female finishers cluster disproportionately in slower performance tiers where the marginal prevalence of pronounced deceleration is higher (Table 2); the crude OR therefore reflects a population-level disparity that is attenuated by this differential demographic mix. The stratified ORs in Table 2 (ranging from 3.32 in the Casual category to 6.14 in the Competitive category) are internally consistent with the adjusted estimate (OR = 3.88) as a category-weighted average of within-stratum effects, all of which exceed the crude OR (2.00) and confirm the direction and magnitude of the negative confounding. We acknowledge that performance category is itself partly determined by sex-related physiological capability and may lie on the causal pathway between sex and pacing failure; we therefore interpret the adjusted OR as the within-category sex effect rather than as a population-level causal contrast, and we report both the crude and the adjusted estimates to make this distinction transparent. Variance inflation factors and model diagnostics (Supplementary Table S6) confirm that multicollinearity between sex, age, and performance category is modest (all VIF < 5).

It is also important to acknowledge the influence of environmental conditions on race dynamics. Recent observational studies of the Berlin Marathon have established that environmental factors are strong predictors of running performance³⁴. Furthermore, specific climatic variables, such as temperature and barometric pressure, have been shown to directly modulate the pacing strategies and running speeds of even the fastest elite competitors³⁵. However, the strength of the current investigation lies in its 27-year longitudinal scope, which encompasses a wide spectrum of weather patterns. The persistence of the sex-based pacing disparity across this extensive timeline suggests that the male tendency toward more aggressive pacing is a pattern that appears robust across the variety of environmental conditions captured in the 27-year dataset.

The fine-grained pacing metrics (within-runner coefficient of variation across 5 km segments, the sustained inflection point, and late-race deceleration) jointly characterize the *shape* of the slowdown rather than only its endpoint magnitude. They show that the sex disparity is not confined to the tail of the slowdown distribution (Figs. 1) but is also expressed in the mean response across performance categories (Fig. 3) and earlier in the race trajectory over a wider segment (Fig. 4). This strengthens the interpretation that the observed difference reflects a systematic contrast in pacing strategy across the race, rather than a discrete late-race failure event affecting a small fraction of male runners.

Practical applications

These findings suggest that male runners—particularly those aiming for aggressive time goals—would benefit from pacing strategies that mimic the female approach: conservative starting splits and a focus on negative splitting. Future coaching interventions could explore strategies addressing the apparent tendency in male runners toward aggressive early pacing. Furthermore, while men possess the physiological advantages (e.g., VO_2max , hemoglobin mass) that dictate the upper limits of absolute speed, women appear to possess superior regulatory mechanisms for energy management. Finally, these findings reinforce the importance of considering biological sex as a key determinant in the interpretation of athletic performance and exercise responses⁷.

Limitations

Several limitations should be acknowledged

First, while the dataset is extensive, it relies on net finish times and splits without direct physiological measures (e.g., heart rate, lactate, or glycogen levels), preventing a definitive causal link between pacing decline and specific metabolic events. The pacing-derived definition of “hitting the wall” is an operational proxy for catastrophic deceleration; direct evidence of substrate exhaustion (e.g., muscle biopsy or indirect calorimetry) was not available, and the metabolic mechanisms discussed should be interpreted as plausible inferences rather than directly measured outcomes.

Second, although the Berlin Marathon course is flat, weather conditions varied across the 27-year period; while our large sample size mitigates individual anomalies, extreme weather years could influence aggregate pacing strategies. The temporal trend analysis (Fig. 5) revealed substantial inter-annual variability in the sex prevalence gap (range 2.8 to 19.9 percentage points), likely reflecting environmental and cohort composition differences across editions; the absence of a systematic trend across the 27-year archive supports the interpretation that the disparity is structural rather than era-specific.

Third, the operational definition of “hitting the wall” as a $\geq 20\%$ slowdown is necessarily a discretization of what is likely a continuous phenomenon. The binary criterion was adopted to enable interpretable risk comparisons between groups, and its robustness was supported by sensitivity analyses at 15% and 25% thresholds.

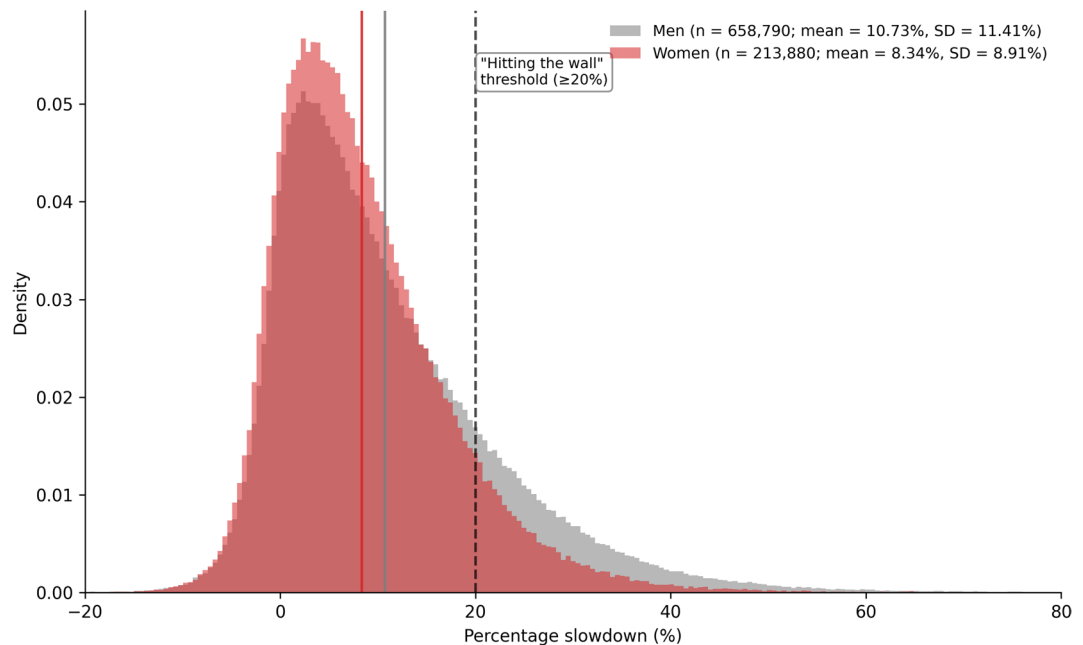


Fig. 1. Probability density distribution of pacing strategies (Percentage Slowdown) by sex among Berlin Marathon finishers (Men: n = 659,294 total, of whom n = 658,790 had valid pacing data plotted; Women: n = 214,040 total, of whom n = 213,880 had valid pacing data plotted). The figure illustrates the kernel density estimation of pacing variability for male (grey) and female (red) runners. The horizontal axis represents the Percentage Slowdown, calculated as the relative time difference between the second and first half of the race, where positive values indicate deceleration (positive split). Visually, the peak of the female distribution is shifted to the left relative to the male peak, indicating that women, on average, maintain a strategy closer to an even split. The male distribution is noticeably wider (greater dispersion) and exhibits a more pronounced right tail (higher skewness), demonstrating a higher relative frequency of extreme deceleration compared to the more compactly clustered female cohort.

(Supplementary Table S2). The graded severity distribution (Supplementary Table S2) further reveals that the sex disparity persists across the continuum of slowdown magnitudes. Future work could explore individualized pacing expectations (e.g., z-score deviations from predicted pacing) or dynamic fatigue modeling to better characterize the gradual nature of pacing collapse¹⁶. More advanced trajectory analyses (changepoint detection, mixed-effects pacing models, functional data analysis of full split trajectories) were not pursued in the present descriptive framework and represent natural extensions for future work.

Fourth, the analytical cohort comprises only those runners who completed the marathon. The official BMW Berlin Marathon results archive, by design, records finishers exclusively; runners who did not finish (DNF) are absent from the dataset and consequently from the present analyses. The true prevalence of catastrophic pacing failure in the broader starting field is therefore likely to be underestimated, since some runners who experience severe pacing collapse may withdraw before the finish line rather than complete the race at substantially reduced pace. This finisher-only constraint is shared with other large-scale archival marathon analyses¹⁶. While we cannot directly quantify the magnitude of this bias, the sex comparison that constitutes the primary inferential target is unlikely to be qualitatively affected unless DNF rates differ substantially between male and female runners—a question that future work integrating start-line registration data with finish results could address.

Fifth, the analytical sample comprises race entries rather than uniquely identified runners; the same individual may complete the marathon in multiple years. Although a sensitivity analysis using a deduplicated subset (Supplementary Table S3) confirmed the direction and approximate magnitude of the sex disparity, the deduplication relied on imperfect name matching (composite key of normalized name + age group) and is therefore conservative; some entries from the same runner may have remained classified as distinct due to spelling variations across editions. We treat the deduplicated analysis as a robustness check rather than a definitive correction; future archival collaborations with race organizers using persistent runner identifiers could provide a more precise estimate of the within-runner variance component.

Finally, although the multivariable logistic regression and within-cohort percentile re-stratification (Supplementary Tables S1 and S4) provide complementary demographic adjustment, the present study did not employ external age-graded performance standards (e.g., World Masters Athletics tables); future work could extend this normalization to triangulate the magnitude of the sex effect under explicit age-graded benchmarks.

An additional methodological consideration concerns one of the fine-grained pacing metrics (oscillation count), where the observed direction was opposite to that initially anticipated (women showed slightly more sign-changes in the pace gradient than men, $d = -0.18$; Supplementary Table S5). The mechanistic interpretation of this pattern requires further investigation and is not advanced in the present work.

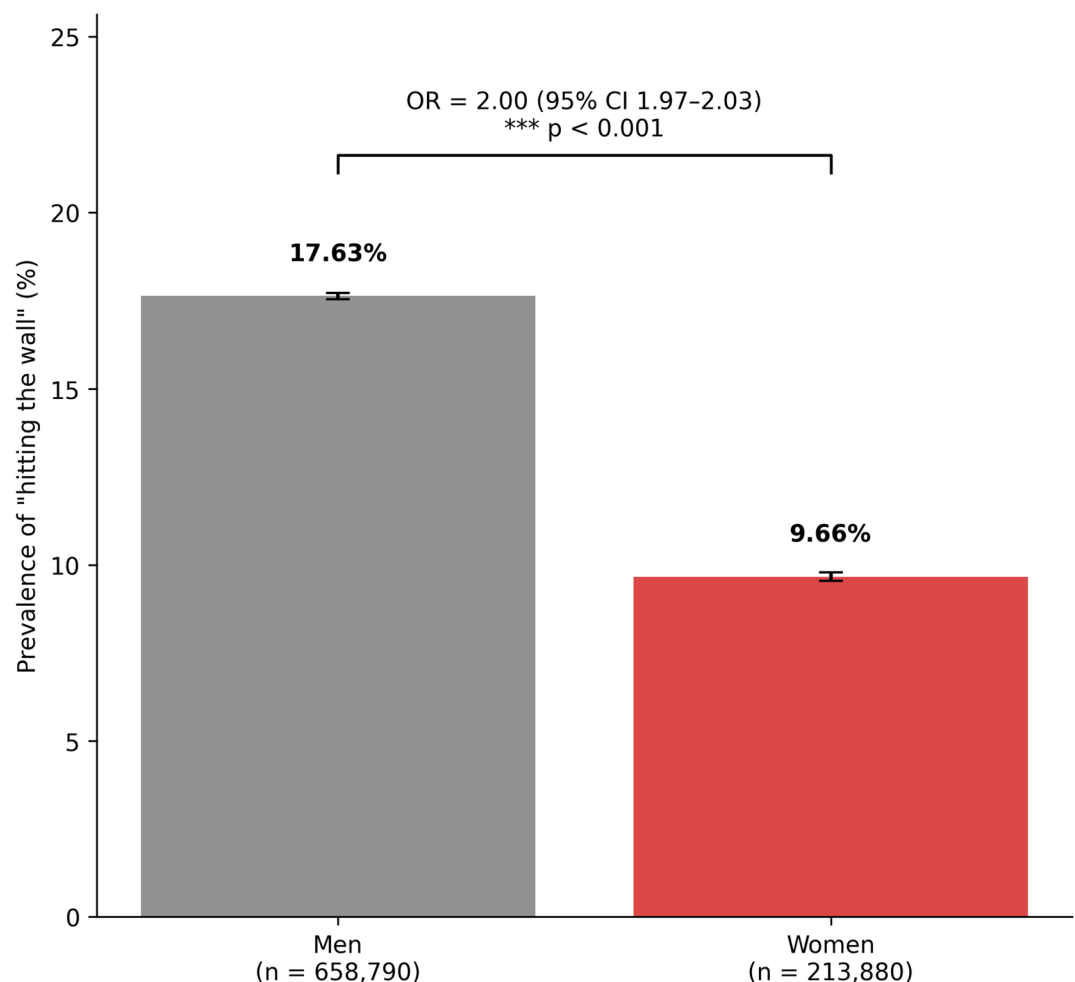


Fig. 2. Prevalence of catastrophic deceleration (“hitting the wall”) by sex among Berlin Marathon finishers. The figure displays the proportion of male (grey) and female (red) runners who experienced a pacing collapse, operationally defined as a percentage slowdown of $\geq 20\%$ in the second half of the race relative to the first. Error bars represent 95% confidence intervals computed via the Wilson score interval for proportions. The data illustrates a marked disparity in pacing failure rates, with male runners being significantly more likely to reach this threshold compared to female runners (17.63% vs. 9.66%, with non-overlapping 95% CIs). This difference corresponds to a crude Odds Ratio (OR) of 2.00 (95% CI 1.97–2.03), indicating that men face a twofold greater relative risk of pronounced second-half deceleration compared to women within the same event conditions.

Conclusions

This analysis of 873,334 Berlin Marathon finishers (1999–2025) provides the largest single-event evidence to date of a substantial sex disparity in pacing collapse, with male runners approximately twice as likely overall—and six times as likely within the sub-3 h cohort—to experience catastrophic deceleration. The persistence of this disparity across all performance tiers, including the most competitive, suggests that experience and physiological capacity alone do not eliminate the gap. These findings are consistent with the hypothesis that behavioral and strategic factors contribute alongside physiological determinants to sex differences in marathon outcomes, and provide a quantitative basis for further investigation of the underlying mechanisms.

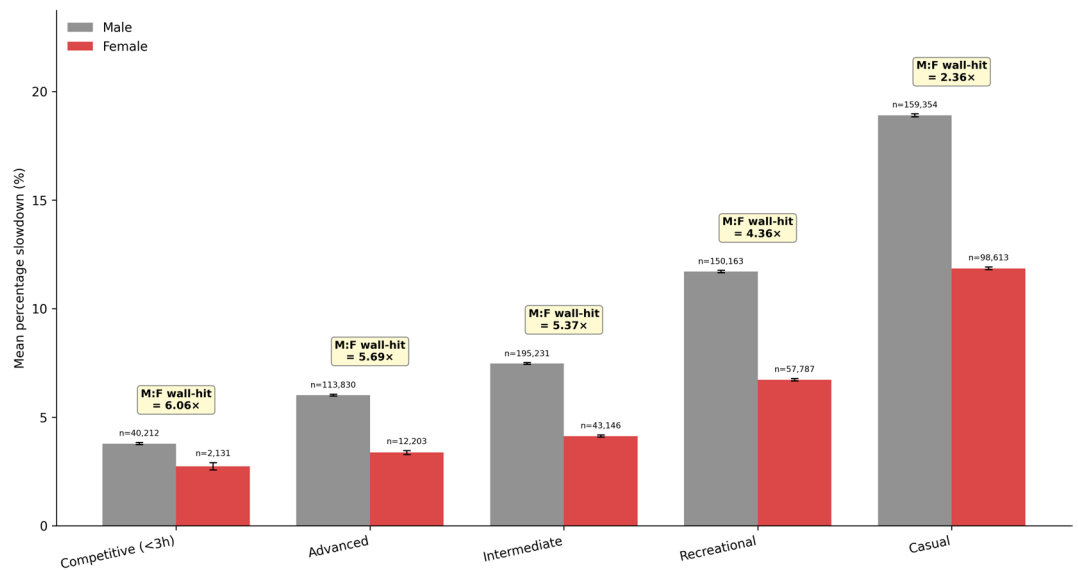


Fig. 3. Sex disparity in mean percentage slowdown stratified by performance level (Finish Time). Comparison of the mean percentage slowdown between male (grey bars) and female (red bars) runners across five distinct performance categories, ranging from Competitive (<3:00 h) to Casual (>4:30 h). Error bars represent the 95% confidence interval of the mean ($1.96 \times$ standard error); sample sizes per category $\times$ sex are indicated above each bar. The data reveals a consistent “pacing gap” where male runners exhibit significantly greater deceleration than female runners within every performance tier ($p < 0.001$). Notably, while the magnitude of the slowdown increases for both sexes as finish times increase, the relative disparity persists even among the fastest cohorts, challenging the assumption that higher performance levels eliminate sex-based pacing differences.

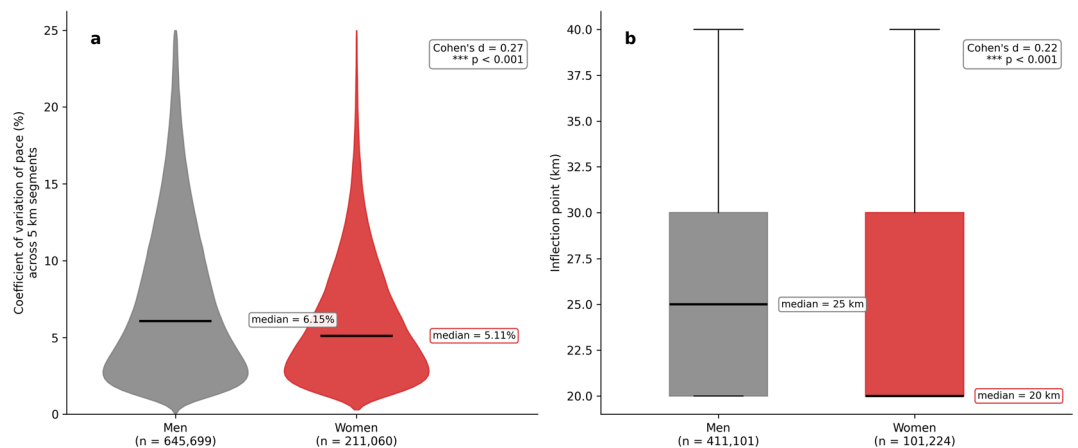


Fig. 4. Fine-grained pacing analysis among runners with valid 5 km splits (Men: $n = 645,699$; Women: $n = 211,060$). **(a)** Violin plot of the coefficient of variation (CV) of pace across 5 km segments by sex, illustrating greater pacing irregularity in male runners (median higher; broader upper tail). **(b)** Boxplot of the inflection point (km), defined as the first 5 km segment with sustained pace deterioration relative to the pre-half-marathon mean, by sex. Boxes show the interquartile range; whiskers extend to $1.5 \times$ IQR; the central line marks the median. Runners without a defined inflection—under the strict criterion that pace must exceed the pre-half-marathon mean by $> 5\%$ and remain slower for all subsequent segments through the finish—represent $\sim 36\%$ of men and $\sim 52\%$ of women and are excluded from the boxplot.

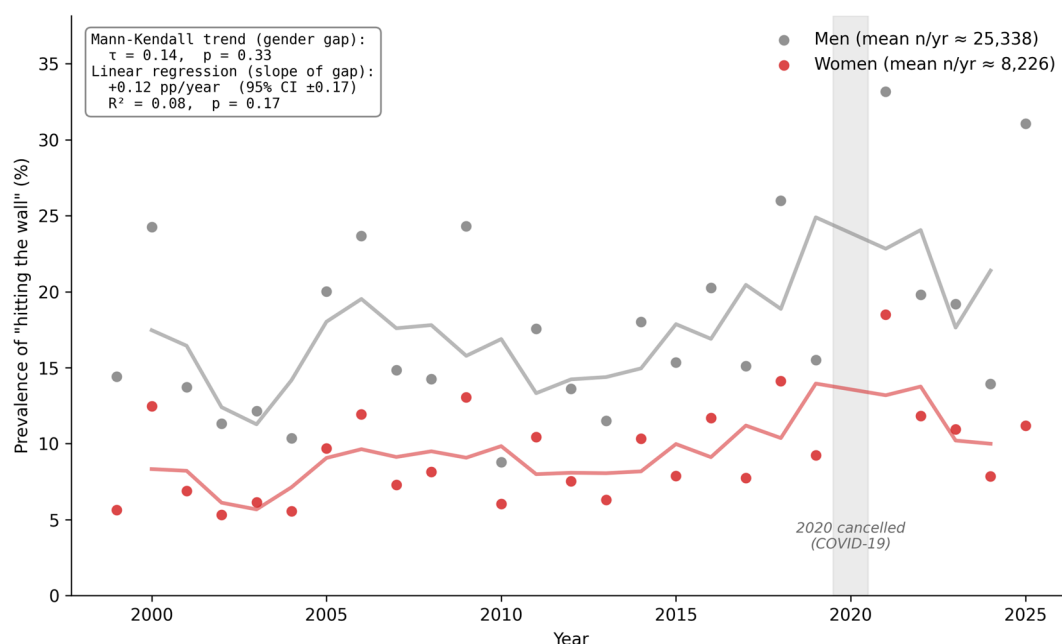


Fig. 5. Temporal evolution of the sex disparity in wall-hit prevalence across the Berlin Marathon archive (1999–2025; 26 editions analyzed; 2020 cancelled due to COVID-19). Annual prevalence by sex (scatter; men grey, women red) with smoothed 3-year rolling-mean trends (lines). The Mann–Kendall trend test indicated no statistically detectable change in the prevalence gap ($\tau = 0.14$, $p = 0.33$); linear regression with year as predictor of the gap likewise did not reach significance (slope = +0.12 percentage points/year, $p = 0.17$). The high inter-annual variability (gap range 2.8–19.9 percentage points) likely reflects environmental and cohort composition differences across editions; the absence of a systematic temporal trend supports the interpretation that the sex disparity is a structural feature of marathon pacing across the 27-year archive.

Data availability

This study employed a retrospective observational design leveraging a large-scale dataset to analyze pacing strategies and performance resilience among marathon runners. Data were extracted from the official BMW Berlin Marathon Results Archive (<https://www.bmw-berlin-marathon.com/en/your-race/results>), covering a 27-year period from 1999 to 2025.

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Author contributions

Aldo Seffrin performed the analyses and drafted the manuscript, Elias Villiger obtained the data, Marilia Santos Andrade, Thomas Rosemann, Katja Weiss and Beat Knechtle helped in drafting the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Given that the data were obtained from a publicly accessible repository and analyzed in an anonymized format, this study was exempt from Institutional Review Board (IRB) approval, in accordance with standard ethical guidelines for the use of public secondary data.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-56334-7>.

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